

Industrial Internship Report on " Smart City Traffic Forecasting Using Machine Learning"

Prepared by

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Executive Summary

This report presents the work completed during the Industrial Internship conducted by upskill Campus and The IoT Academy in collaboration with UniConverge Technologies Pvt. Ltd. (UCT). During this internship, I developed a Machine Learning-based Smart City Traffic Forecasting System that predicts traffic volume at different city junctions using historical traffic data.

The project involved data preprocessing, feature engineering, model training, evaluation, and deployment using Python and Streamlit. A Random Forest Regressor model was trained to forecast traffic volume with high accuracy.

The developed application allows users to select date, time, and junction information to obtain predicted traffic volume instantly through a user-friendly interface.

This internship enhanced my practical knowledge of Machine Learning, Data Science, Python programming, data visualization, and web application development. It also improved my problem-solving ability and provided valuable experience in developing an industry-oriented project.

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1 Preface

This report summarizes the work completed during my Industrial Internship conducted by **Upskill Campus, The IoT Academy, and UniConverge Technologies Pvt. Ltd. (UCT)**. The internship provided an opportunity to gain practical knowledge in Machine Learning, Python programming, Data Science, and real-world software development.

The primary objective of this internship was to develop a **Smart City Traffic Forecasting Using Machine Learning** system capable of predicting traffic volume based on historical traffic data. The project involved data preprocessing, feature engineering, model training, evaluation, and deployment using Streamlit.

The internship followed a structured learning approach consisting of understanding the problem statement, designing the solution, implementing the project, testing the developed application, and documenting the overall work. Throughout the internship, I improved my technical skills and gained practical experience in developing an end-to-end Machine Learning application.

This internship enhanced my understanding of data analysis, predictive modeling, and web application development while strengthening my problem-solving, analytical thinking, and project implementation skills.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



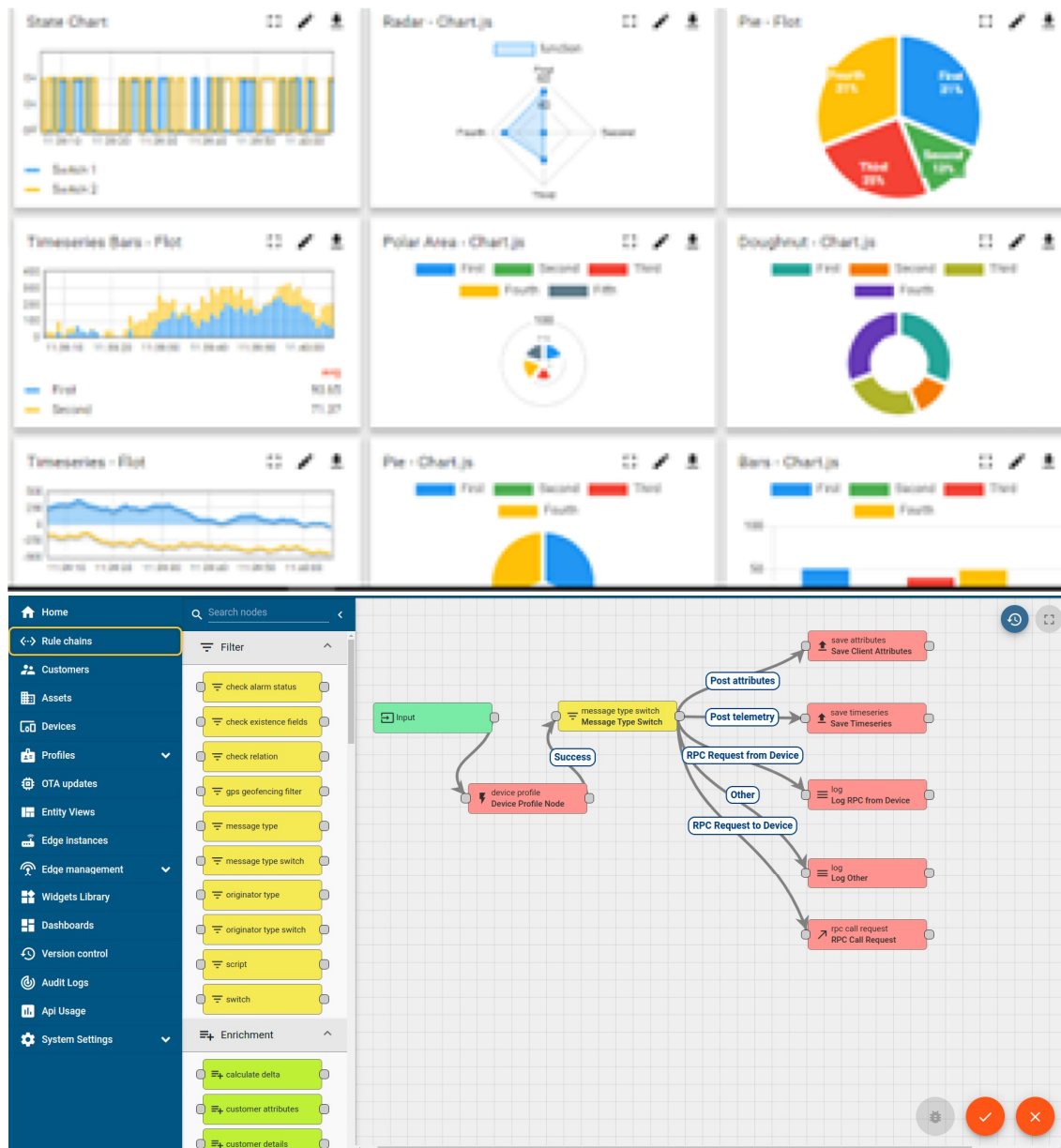
i. UCT IoT Platform ()

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



FACTORY WATCH

ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i



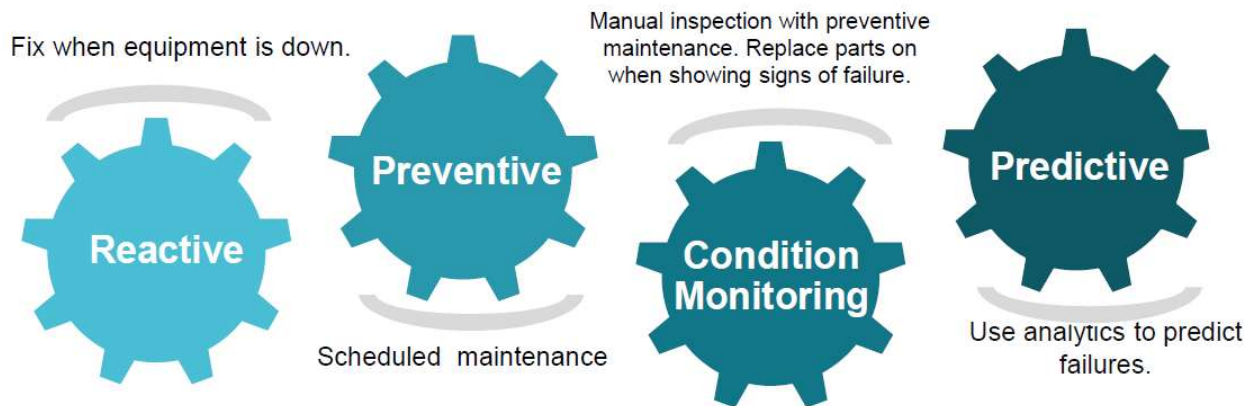


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

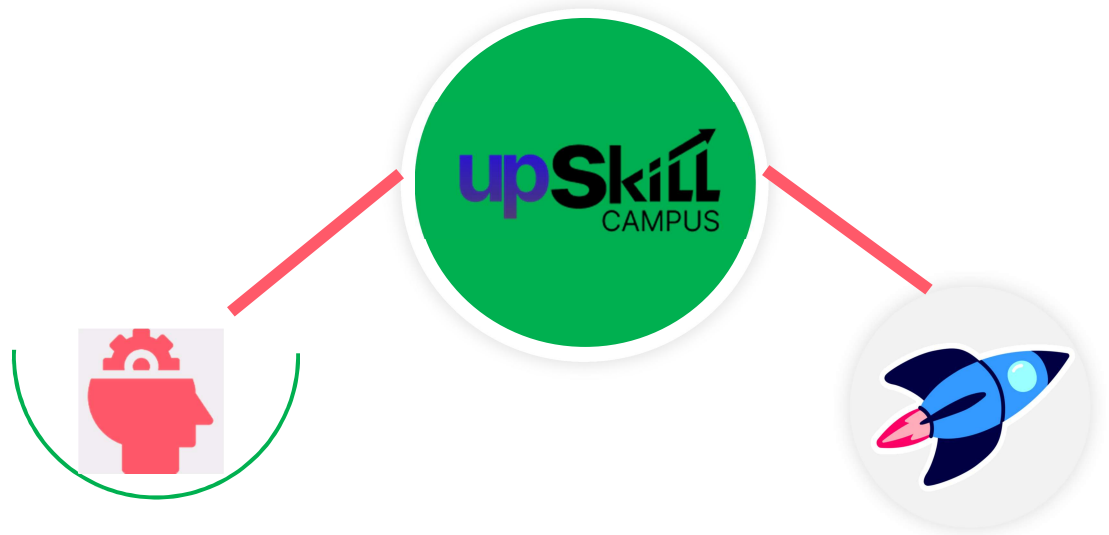
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

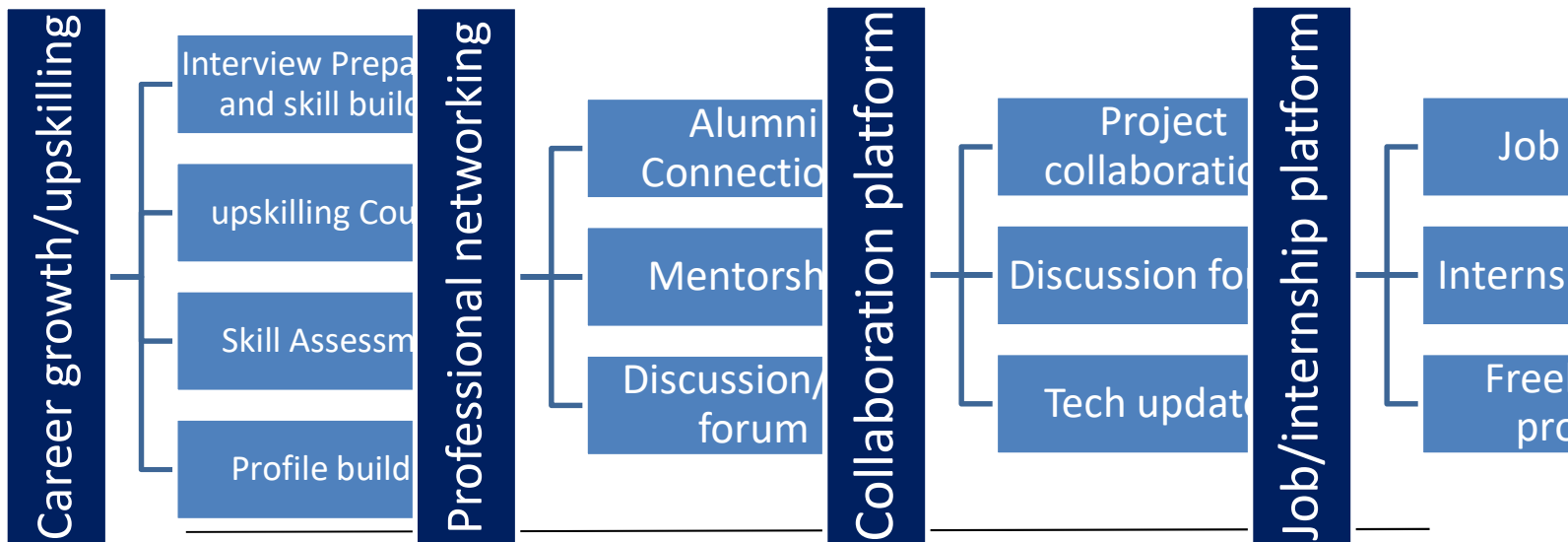
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

<https://www.upskillcampus.com/>



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- get practical experience of working in the industry.
- to solve real world problems.
- to have improved job prospects.
- to have Improved understanding of our field and its applications.
- to have Personal growth like better communication and problem solving.

2.5 Reference

- [1] Scikit-learn Documentation – <https://scikit-learn.org/>
- [2] Streamlit Documentation – <https://streamlit.io/>
- [3] Pandas Documentation – <https://pandas.pydata.org/>

2.6 Glossary

Terms	Acronym
Machine Learning	ML
Random Forest Regressor	RFR
Artificial Intelligence	AI
Mean Absolute Error	MAE
Mean Squared Error	MSE

3 Problem Statement

Traffic congestion has become one of the major challenges in modern cities due to the increasing number of vehicles and rapid urbanization. Conventional traffic management systems mainly rely on fixed traffic signal timings and manual monitoring, which are often unable to adapt to changing traffic conditions.

The objective of this project is to develop a **Smart City Traffic Forecasting System** using Machine Learning that predicts future traffic volume based on historical traffic data. The system analyzes features such as junction number, date, day, month, hour, and day of the week to forecast traffic volume accurately. This prediction helps improve traffic management, reduce congestion, and support better decision-making for smart cities.

4 Existing and Proposed solution

4.1 Existing Solution

Existing traffic management systems mainly depend on manual observation, CCTV monitoring, or fixed traffic signal timings. These approaches provide limited prediction capabilities and cannot accurately forecast future traffic conditions.

4.2 Proposed Solution

The proposed solution is a **Machine Learning-based Smart City Traffic Forecasting System** developed using Python and Streamlit. A **Random Forest Regressor** model is trained using historical traffic data to predict traffic volume based on user inputs such as junction number, date, month, day, hour, and day of the week.

4.3 Value Addition

- Predicts future traffic volume with high accuracy.
- Helps optimize traffic management.
- Supports smart city planning.
- Easy-to-use Streamlit web application.
- Reduces traffic congestion through predictive analysis.

4.4 Code submission (Github link)

<https://github.com/mahek786ssh-ctrl/upskillcampus>

4.5 Report submission (Github link) :

Report GitHub link will be updated after PDF upload.

5 Proposed Design/ Model

Given more details about design flow of your solution. This is applicable for all domains. DS/ML Students can cover it after they have their algorithm implementation. There is always a start, intermediate stages and then final outcome.

5.1 High Level Diagram

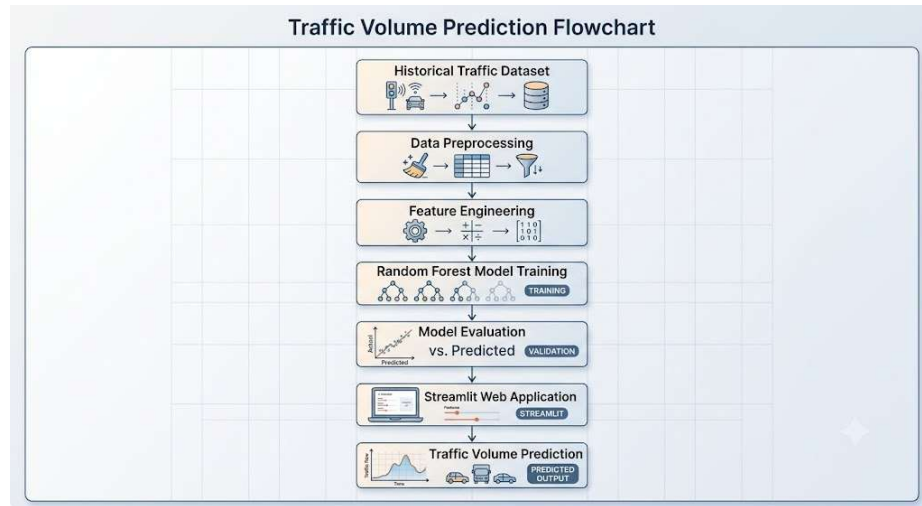


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram

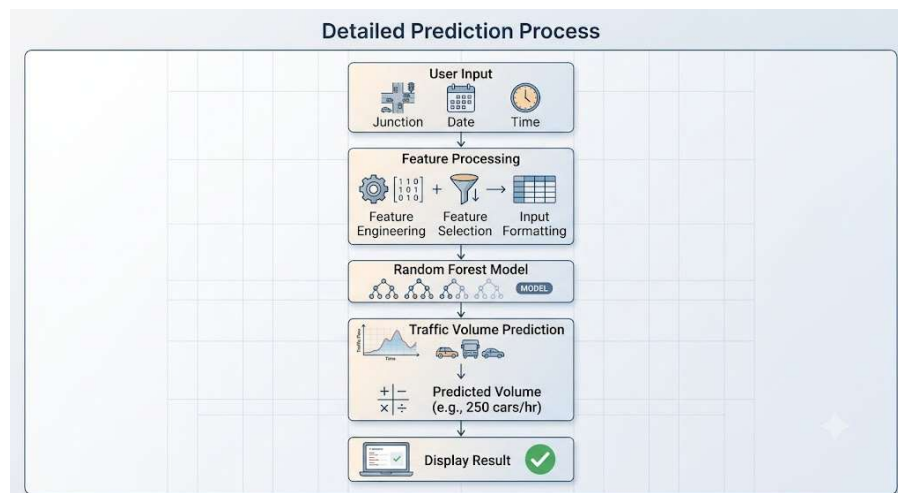


Figure 2: LOW LEVEL DIAGRAM OF THE SYSTEM

5.3 Interfaces

The application provides a user-friendly Streamlit interface where users enter the junction number, date, month, day, hour, and day of the week. The trained Random Forest model processes these inputs and predicts the expected traffic volume, which is displayed instantly.

6 Performance Test

6.1 Test Plan/ Test Cases

Test Case	Expected Result	Status
Valid user input	Accurate traffic prediction	Pass
Invalid input	Error handling	Pass
Different junction selection	Correct prediction	Pass
Multiple predictions	Successful execution	Pass

6.2 Test Procedure

- Collected and preprocessed historical traffic data.
- Trained the Random Forest Regressor model.
- Evaluated the model using MAE, MSE, and R^2 Score.
- Tested the Streamlit application using different traffic scenarios.

6.3 Performance Outcome

The developed Smart City Traffic Forecasting System achieved:

- **Model:** Random Forest Regressor
- **Mean Absolute Error (MAE):** 2.3987
- **Mean Squared Error (MSE):** 12.6289
- **R^2 Score:** 0.9690 (96.9% Accuracy)

The model successfully predicts traffic volume with high accuracy and demonstrates reliable performance for smart traffic forecasting.

7 My learnings

During this internship, I gained practical knowledge of Machine Learning, Python programming, Data Science, and Streamlit application development. I learned how to preprocess datasets, perform feature engineering, train and evaluate machine learning models, and deploy applications through a user-friendly web interface.

I also improved my problem-solving, analytical thinking, debugging, documentation, and project implementation skills. This internship helped me understand the complete Machine Learning development lifecycle and strengthened my confidence in building real-world AI applications.

8 Future work scope

The current system can be further enhanced by integrating real-time traffic data from IoT sensors and GPS devices. Future improvements may include deep learning models for higher prediction accuracy, weather and event-based traffic forecasting, live traffic visualization, and cloud deployment for large-scale smart city applications.

The project can also be integrated with intelligent traffic signal systems to automatically optimize signal timings based on predicted traffic conditions, making urban transportation more efficient and sustainable.